

Student Guide — senior / module-12- movement-of-sun

IKS Curriculum

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Student Workbook

Student Workbook — Movement of the Sun (Senior)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Sūrya* — _____
- *Uttarāyana* — _____
- *Dakṣiṇāyana* — _____
- *saṁkrānti* — _____
- *rāśi* — _____
- *sāyana* — _____
- *nirayana* — _____

Day-by-day notes

Day 1 — The Sun's Yearly Path

What I learned:

One question I have:

Day 2 — Axial Tilt and the Seasons

What I learned:

One question I have:

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

What I learned:

One question I have:

Day 4 — Equinoxes — Viṣuva

What I learned:

One question I have:

Day 5 — Twelve Rāśis

What I learned:

One question I have:

Day 6 — Saṁkrānti Moments

What I learned:

One question I have:

Day 7 — Sāyana vs Nirayana

What I learned:

One question I have:

Day 8 — Precession of the Equinoxes

What I learned:

One question I have:

Day 9 — Cross-Cultural Solar Calendars

What I learned:

One question I have:

Day 10 — Final Share and Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Sun’s Yearly Path

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: observe that sunrise direction changes; introduce the ecliptic.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)

Primary source (today’s reading)

From *Sūrya-siddhānta* (c. 5th–10th c. CE) — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces The Sun’s Yearly Path.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — Axial Tilt and the Seasons

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: demonstrate with torch+globe why we have summer and winter.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *Uttarāyaṇa* (IAST: *Uttarāyaṇa*; lit. “northward solar course (Dec 21 – June 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Axial Tilt and the Seasons.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify the dates and direction of each.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Dakṣiṇāyana* (IAST: Dakṣiṇāyana; lit. “southward solar course (June 21 – Dec 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Uttarāyana and Dakṣiṇāyana.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — Equinoxes — Viṣuva

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify spring and autumn equinoxes (day = night).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *saṁkrānti* (IAST: saṁkrānti; lit. “Sun’s crossing from one Rāśi to the next”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Equinoxes — Viṣuva.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 5 — Twelve Rāśis

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: name the 12 rāśis in sanskrit and pair with greek zodiac names.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *rāśi* (IAST: rāśi; lit. “1/12th of the ecliptic; zodiacal sign”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Twelve Rāsis.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Saṃkrānti Moments

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify makara, karka, and other saṃkrāntis on the calendar.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *sāyana* (IAST: *sāyana*; lit. “tropical zodiac (linked to seasons)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Saṁkrānti Moments.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.

Materials

- Whiteboard / chart paper
- Notebook, pen per student

- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *nirayana* (IAST: nirayana; lit. “sidereal zodiac (linked to fixed stars)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Sāyana vs Nirayana.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 8 — Precession of the Equinoxes

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: explain why indian and western zodiac dates differ today.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Precession of the Equinoxes.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 9 — Cross-Cultural Solar Calendars

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compare with egyptian, roman, gregorian.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Uttarāyaṇa* (IAST: *Uttarāyaṇa*; lit. “northward solar course (Dec 21 – June 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Cross-Cultural Solar Calendars.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Final Share and Assessment

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: project share + summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *Dakṣiṇāyana* (IAST: *Dakṣiṇāyana*; lit. “southward solar course (June 21 – Dec 21)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Final Share and Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Writing Prompts

Writing Prompts — Movement of the Sun (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Movement of the Sun to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Movement of the Sun with a similar concept from another module or from your science textbook. What's similar? Different?

4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Movement of the Sun? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Movement of the Sun (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sun's Yearly Path** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Axial Tilt and the Seasons** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Uttarāyaṇa and Dakṣiṇāyana** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Equinoxes — Viṣuva** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Twelve Rāsis** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Saṁkrānti Moments** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Sāyana vs Nirayana** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Precession of the Equinoxes** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Cross-Cultural Solar Calendars** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.

- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.